

Network Infrastructure

Multi-Building Enterprise Network — Design & Implementation Report

Prepared by

Mahmoud Walid Said Elhalawany

Course: Network Infrastructure

Project File (Cisco Packet Tracer): bit.ly/MahmoudWalidElhalawany

Introduction

في هذا المشروع تم تصميم شبكة تتكون من ثلاثة مباني، كل مبنى يحتوي على ثلاثة طوابق. كل طابق متصل بـ Access Switch خاص به، وجميع الـ Access Switches متصلة بـ Multilayer Switch داخل نفس المبنى، حيث يقوم الـ Multilayer Switch بالتوجيه (Routing) بين الـ VLANs والشبكات الفرعية (Subnets) الخاصة بالطوابق الثلاثة، مما يتيح للأجهزة داخل المبنى التواصل مع بعضها البعض.

أما التواصل بين المباني المختلفة، فيتم من خلال Router Edge، حيث إن كل Multilayer Switch متصل بالـ Router Edge، وهو دوره مسؤول عن:

- تحقيق الاتصال بين المباني المختلفة (Inter-Building Communication).
- توصيل الشبكة الداخلية بمزود خدمة الإنترنت (ISP – Internet Service Provider).
- تطبيق NAT لترجمة العناوين الخاصة الداخلية إلى عنوان عام (Public IP).

بهذا التصميم:

- يتم تحقيق الاتصال الداخلي بين الطوابق عبر الـ Multilayer Switch.
- يتم تحقيق الاتصال بين المباني والخروج إلى الإنترنت عبر الـ Router Edge.
- تتوفر خدمات الشبكة مثل Web, Email, FTP, DNS, DHCP، بالإضافة إلى تطبيق سياسات الحماية (Security Port, ACLs).

Building 1

Floor 1

Number of Hosts	16 PCs
VLAN	VLAN 10
Subnet	10.0.0.0/27
Gateway (Virtual IP)	10.0.0.1 (virtual — HSRP Active/Standby on Multilayer Switch)
Broadcast Address	10.0.0.31
DHCP	Assigned by the Multilayer Switch (Active)
Usable IP Addresses	30
Used IPs	16 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 19
Available IPs (not used)	11

PC0 is assigned a static IP — 10.0.0.4. Reason: this device is the target for an Access Control List (ACL) test case.

Access Switch — Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports)

Interface Range	Mode	Purpose
Fa0/3 – Fa0/18	Access	Connects PCs — assigned to VLAN 10
Fa0/1 – Fa0/2	Trunk	Uplink to the Multilayer Switch
Fa0/19 – Fa0/24, Gi0/1 – Gi0/2	Unused	Reserved for future expansion

Interface Fa0/5 is reserved for Port Security (Sticky MAC):

```
interface FastEthernet0/5
  switchport mode access
  switchport port-security
  switchport port-security maximum 1
  switchport port-security mac-address sticky
  switchport port-security violation shutdown
```

Floor 2

Number of Hosts	10 PCs
VLAN	VLAN 20
Subnet	10.0.0.96/28
Gateway (Virtual IP)	10.0.0.97 (virtual — HSRP Active/Standby on Multilayer Switch)
Broadcast Address	10.0.0.111

DHCP	Assigned by the Multilayer Switch (Active)
Usable IP Addresses	14
Used IPs	10 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 13
Available IPs (not used)	1

Access Switch — Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports)

Interface Range	Mode	Purpose
Fa0/3 – Fa0/12	Access	Connects PCs — assigned to VLAN 20
Fa0/1 – Fa0/2	Trunk	Uplink to the Multilayer Switch
Fa0/13 – Fa0/24, Gi0/1 – Gi0/2	Unused	Reserved for future expansion

Floor 3

Number of Hosts	12 PCs
VLAN	VLAN 30
Subnet	10.0.0.64/27
Gateway (Virtual IP)	10.0.0.65 (virtual — HSRP Active/Standby on Multilayer Switch)
Broadcast Address	10.0.0.95
DHCP	Assigned by the Multilayer Switch (Active)
Usable IP Addresses	30
Used IPs	12 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 15
Available IPs (not used)	15

Access Switch — Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports)

Interface Range	Mode	Purpose
Fa0/3 – Fa0/14	Access	Connects PCs — assigned to VLAN 30
Fa0/1 – Fa0/2	Trunk	Uplink to the Multilayer Switch
Fa0/15 – Fa0/24, Gi0/1 – Gi0/2	Unused	Reserved for future expansion

Multilayer Switch 0 (Active)

Model	Cisco Catalyst 6500 (Multilayer Switch)
Role	Active switch in HSRP — provides routing between the floors of Building 1 and forwards traffic to the Edge Router.

VLANs and Interfaces

VLAN 10 (Floor 1)	Interface IP 10.0.0.2/27 · Virtual IP (HSRP) 10.0.0.1 — default gateway for Floor 1 PCs
VLAN 20 (Floor 2)	Interface IP 10.0.0.98/28 · Virtual IP (HSRP) 10.0.0.97 — default gateway for Floor 2 PCs
VLAN 30 (Floor 3)	Interface IP 10.0.0.66/27 · Virtual IP (HSRP) 10.0.0.65 — default gateway for Floor 3 PCs

Trunk Interfaces

Gi1/0/1 – Gi1/0/3 configured as trunk ports, connecting the Multilayer Switch to the access switches of the three floors.

DHCP Configuration

Pool F1 (Floor 1)	Network 10.0.0.0/27
Default Router	10.0.0.1
DNS	10.0.0.180
Excluded Addresses	10.0.0.1 – 10.0.0.4

Pool F2 (Floor 2)	Network 10.0.0.96/28
Default Router	10.0.0.97
DNS	10.0.0.180
Excluded Addresses	10.0.0.97 – 10.0.0.99

Pool F3 (Floor 3)	Network 10.0.0.64/27
Default Router	10.0.0.65
DNS	10.0.0.180
Excluded Addresses	10.0.0.65 – 10.0.0.67

Certain addresses were excluded from each pool to reserve them for static assignment. Example — VLAN 10 (Floor 1): 10.0.0.1 = virtual gateway (HSRP), 10.0.0.2 = Active MLS interface, 10.0.0.3 = Standby MLS interface, 10.0.0.4 = reserved for PC0 (static, used in ACL testing).

Uplink to Edge Router

Interface Gi1/0/4 is configured as a routed port with IP address 10.0.0.185/30, connected directly to the Edge Router.

HSRP Configuration

- This Multilayer Switch is configured as the Active HSRP device for VLANs 10, 20, and 30.
- The second Multilayer Switch in Building 1 acts as the Standby HSRP device.
- Priority is set to 150 (higher than the default 100), ensuring this switch always takes the Active role when available.
- Preempt is enabled — if the Active switch fails and later recovers, it automatically reclaims the Active role.

Routing

- Default route configured to forward unknown traffic to the Edge Router (next-hop 10.0.0.186).
- OSPF enabled for internal routing between VLANs and to share routes with the Edge Router, ensuring connectivity between buildings.

Remote Access (Telnet)

Telnet is enabled on the HSRP virtual IP for each VLAN. When connecting via Telnet to the virtual IP (e.g. 10.0.0.1 for VLAN 10), the session is always established with the Active Multilayer Switch — allowing quick verification of which switch currently holds the Active role.

Multilayer Switch 1 (Standby)

Model	Cisco Catalyst 6500 (Multilayer Switch)
Role	Standby switch in HSRP — provides redundancy for the Active switch and takes over only if it fails.

VLANs and Interfaces

VLAN 10 (Floor 1)	Interface IP 10.0.0.3/27 · Virtual IP (HSRP) 10.0.0.1 — default gateway for Floor 1 PCs
VLAN 20 (Floor 2)	Interface IP 10.0.0.99/28 · Virtual IP (HSRP) 10.0.0.97 — default gateway for Floor 2 PCs
VLAN 30 (Floor 3)	Interface IP 10.0.0.67/27 · Virtual IP (HSRP) 10.0.0.65 — default gateway for Floor 3 PCs

Trunk Interfaces

Gi1/0/1 – Gi1/0/3 configured as trunk ports, connecting the Multilayer Switch to the access switches of the three floors.

DHCP Configuration

Pool F1 (Floor 1)	Network 10.0.0.0/27
Default Router	10.0.0.1
DNS	10.0.0.180
Excluded Addresses	10.0.0.1 – 10.0.0.4 (Virtual IP + Active + Standby + static PC0)

Pool F2 (Floor 2)	Network 10.0.0.96/28
Default Router	10.0.0.97
DNS	10.0.0.180
Excluded Addresses	10.0.0.97 – 10.0.0.99 (Virtual IP + Active + Standby)

Pool F3 (Floor 3)	Network 10.0.0.64/27
Default Router	10.0.0.65
DNS	10.0.0.180
Excluded Addresses	10.0.0.65 – 10.0.0.67 (Virtual IP + Active + Standby)

Uplink to Edge Router

Interface Gi1/0/4 is configured as a routed port with IP address 10.0.0.189/30, connected directly to the Edge Router.

HSRP Configuration

- This Multilayer Switch operates as the Standby HSRP device for VLANs 10, 20, and 30.
- No priority or preempt configured by default — it takes over only if the Active switch fails, and does not reclaim the Active role once the Active switch comes back online.

Routing

- Default route configured to forward unknown traffic to the Edge Router (next-hop 10.0.0.190).
- OSPF enabled to advertise VLAN networks and learn routes from the Edge Router, ensuring inter-building connectivity.

Remote Access (Telnet)

Telnet is configured, but sessions through the virtual IP are always directed to the Active switch.

Edge Router 0

Model	PT-Router (Packet Tracer Default Router)
Role	Connects the internal network of Building 1 to the ISP, applies NAT, and enforces ACL policies. It also acts as the interconnection point between buildings, ensuring devices in Building 1, Building 2, and Building 3 can communicate through routing.

Interfaces and IP Addresses

Interface	IP Address	Connects To
FastEthernet 0/0	10.0.0.186/30	Active Multilayer Switch

Interface	IP Address	Connects To
FastEthernet 1/0	10.0.0.190/30	Standby Multilayer Switch
GigabitEthernet 9/0	20.0.0.1/30	ISP (public-facing interface)
Serial 2/0	10.0.0.209/30	Edge Router of Building 2

Routing

- Default route: 0.0.0.0/0 → next-hop 20.0.0.2 (ISP).
- OSPF configured to advertise the routed links to the Multilayer Switches, allowing all internal subnets of the three floors to reach the router, and ensuring inter-building connectivity as well as access to the ISP.

Advertised Networks (OSPF)

- 10.0.0.184/30 — link to Active MLS
- 10.0.0.188/30 — link to Standby MLS
- 10.0.0.208/30 — backbone / inter-building link to Building 2

Access Control List (ACL)

- Access-list 110 denies TCP traffic from host 10.0.0.4 (PC0, static IP) to any FTP service (port 21).
- All other IP traffic is permitted.
- Applied inbound on interface FastEthernet 0/0.

```
access-list 110 deny tcp host 10.0.0.4 any eq ftp
access-list 110 permit ip any any

interface FastEthernet0/0
 ip access-group 110 in
```

Network Address Translation (NAT)

Type	NAT Overload (PAT)
NAT Inside Interfaces	FastEthernet 0/0, FastEthernet 1/0
NAT Outside Interface	GigabitEthernet 9/0
Translation	All inside networks share the public IP of Gi9/0 (20.0.0.1) using overload

Inside Subnets Permitted for NAT (ACL 1):

- 10.0.0.0/27 (Floor 1)
- 10.0.0.96/28 (Floor 2)
- 10.0.0.64/27 (Floor 3)

```
access-list 1 permit 10.0.0.0 0.0.0.31
```

```
access-list 1 permit 10.0.0.96 0.0.0.15
access-list 1 permit 10.0.0.64 0.0.0.31
ip nat inside source list 1 interface GigabitEthernet9/0 overload

interface GigabitEthernet9/0
 ip nat outside
interface FastEthernet0/0
 ip nat inside
interface FastEthernet1/0
 ip nat inside
```

Building 2

Floor 1

Number of Hosts	7 PCs
VLAN	VLAN 40
Subnet	10.0.0.144/28
Gateway (Virtual IP)	10.0.0.145 (virtual — HSRP Active/Standby on Multilayer Switch)
Broadcast Address	10.0.0.159
DHCP	Assigned by the Multilayer Switch (Active)
Usable IP Addresses	14
Used IPs	7 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 10
Available IPs (not used)	4

Access Switch — Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports)

Interface Range	Mode	Purpose
Fa0/3 – Fa0/9	Access	Connects PCs — assigned to VLAN 40
Fa0/1 – Fa0/2	Trunk	Uplink to the Multilayer Switch
Fa0/10 – Fa0/24, Gi0/1 – Gi0/2	Unused	Reserved for future expansion

Floor 2 — Server Farm

Number of Hosts	1 PC configured as a multi-service Server
VLAN	VLAN 50
Subnet	10.0.0.176/29
Gateway (Virtual IP)	10.0.0.177 (virtual — HSRP Active/Standby on Multilayer Switch)
Broadcast Address	10.0.0.183
Static IP	10.0.0.180
Usable IP Addresses	6
Used IPs	1 PC + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 4
Available IPs (not used)	2

Services Enabled:

- Web Server (HTTP) — reachable by all clients.
- DNS Server — domain name: www.mahmoud.com.
- Email Server (SMTP/POP3) — configured with user accounts.

- FTP Server — requires authentication for access.

User Accounts (FTP / Email):

- Username: ali · Password: 12345
- Username: noor · Password: 12345

Access Switch — Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports)

Interface Range	Mode	Purpose
Fa0/3	Access	Connects PCs — assigned to VLAN 50
Fa0/1 – Fa0/2	Trunk	Uplink to the Multilayer Switch
Fa0/4 – Fa0/24, Gi0/1 – Gi0/2	Unused	Reserved for future expansion

Floor 3

Number of Hosts	5 PCs
VLAN	VLAN 60
Subnet	10.0.0.160/28
Gateway (Virtual IP)	10.0.0.161 (virtual — HSRP Active/Standby on Multilayer Switch)
Broadcast Address	10.0.0.175
DHCP	Assigned by the Multilayer Switch (Active)
Usable IP Addresses	14
Used IPs	5 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 8
Available IPs (not used)	6

Access Switch — Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports)

Interface Range	Mode	Purpose
Fa0/3 – Fa0/7	Access	Connects PCs — assigned to VLAN 60
Fa0/1 – Fa0/2	Trunk	Uplink to the Multilayer Switch
Fa0/8 – Fa0/24, Gi0/1 – Gi0/2	Unused	Reserved for future expansion

Multilayer Switch 2 (Active)

Model	Cisco Catalyst 6500 (Multilayer Switch)
Role	Active switch in HSRP — provides routing between the floors of Building 2 and forwards traffic to the Edge Router.

VLANs and Interfaces

VLAN 40 (Floor 1)	Interface IP 10.0.0.146/28 · Virtual IP (HSRP) 10.0.0.145 — default gateway for Floor 1 PCs
VLAN 50 (Floor 2)	Interface IP 10.0.0.178/29 · Virtual IP (HSRP) 10.0.0.177 — default gateway for Floor 2 PCs
VLAN 60 (Floor 3)	Interface IP 10.0.0.162/28 · Virtual IP (HSRP) 10.0.0.161 — default gateway for Floor 3 PCs

Trunk Interfaces

Gi1/0/1 – Gi1/0/3 configured as trunk ports, connecting the Multilayer Switch to the access switches of the three floors.

DHCP Configuration

Pool F4 (Floor 1)	Network 10.0.0.144/28
Default Router	10.0.0.145
DNS	10.0.0.180
Excluded Addresses	10.0.0.145 – 10.0.0.147

Pool F6 (Floor 3)	Network 10.0.0.160/28
Default Router	10.0.0.161
DNS	10.0.0.180
Excluded Addresses	10.0.0.161 – 10.0.0.163

Floor 2 hosts the server with a fixed static IP (10.0.0.180), so no DHCP pool is required for VLAN 50.

Uplink to Edge Router

Interface Gi1/0/4 is configured as a routed port with IP address 10.0.0.193/30, connected directly to the Edge Router.

HSRP Configuration

- This Multilayer Switch is configured as the Active HSRP device for VLANs 40, 50, and 60.
- The second Multilayer Switch in Building 2 acts as the Standby HSRP device.
- Priority is set to 150 (higher than the default 100), ensuring this switch always takes the Active role when available.
- Preempt is enabled — if the Active switch fails and later recovers, it automatically reclaims the Active role.

Routing

- Default route configured to forward unknown traffic to the Edge Router (next-hop 10.0.0.194).
- OSPF enabled for internal routing between VLANs and to share routes with the Edge Router, ensuring connectivity between buildings.

Multilayer Switch 3 (Standby)

Model	Cisco Catalyst 6500 (Multilayer Switch)
Role	Standby switch in HSRP — provides redundancy for the Active switch and takes over only if it fails.

VLANs and Interfaces

VLAN 40 (Floor 1)	Interface IP 10.0.0.147/28 · Virtual IP (HSRP) 10.0.0.145 — default gateway for Floor 1 PCs
VLAN 50 (Floor 2)	Interface IP 10.0.0.179/29 · Virtual IP (HSRP) 10.0.0.177 — default gateway for Floor 2 PCs
VLAN 60 (Floor 3)	Interface IP 10.0.0.163/28 · Virtual IP (HSRP) 10.0.0.161 — default gateway for Floor 3 PCs

Trunk Interfaces

Gi1/0/1 – Gi1/0/3 configured as trunk ports, connecting the Multilayer Switch to the access switches of the three floors.

DHCP Configuration

Pool F4 (Floor 1)	Network 10.0.0.144/28
Default Router	10.0.0.145
DNS	10.0.0.180
Excluded Addresses	10.0.0.145 – 10.0.0.147

Pool F6 (Floor 3)	Network 10.0.0.160/28
Default Router	10.0.0.161
DNS	10.0.0.180
Excluded Addresses	10.0.0.161 – 10.0.0.163

Uplink to Edge Router

Interface Gi1/0/4 is configured as a routed port with IP address 10.0.0.197/30, connected directly to the Edge Router.

HSRP Configuration

- This Multilayer Switch operates as the Standby HSRP device for VLANs 40, 50, and 60.
- No priority or preempt configured by default — it takes over only if the Active switch fails, and does not reclaim the Active role once the Active switch comes back online.

Routing

- Default route configured to forward unknown traffic to the Edge Router (next-hop 10.0.0.198).
- OSPF enabled to advertise VLAN networks and learn routes from the Edge Router, ensuring inter-building connectivity.

Edge Router 1

Model	PT-Router (Packet Tracer Default Router)
Role	Connects the internal network of Building 2 to the ISP and applies NAT. It also acts as the interconnection point between buildings, ensuring devices in Building 1, Building 2, and Building 3 can communicate through routing.

Interfaces and IP Addresses

Interface	IP Address	Connects To
FastEthernet 0/0	10.0.0.194/30	Active Multilayer Switch
FastEthernet 1/0	10.0.0.198/30	Standby Multilayer Switch
GigabitEthernet 9/0	20.0.0.5/30	ISP (public-facing interface)
Serial 2/0	10.0.0.210/30	Edge Router of Building 1
Serial 3/0	10.0.0.213/30	Edge Router of Building 3

Routing

- Default route: 0.0.0.0/0 → next-hop 20.0.0.6 (ISP).
- OSPF configured to advertise the routed links to the Multilayer Switches, allowing all internal subnets of the three floors to reach the router, and ensuring inter-building connectivity as well as access to the ISP.

Advertised Networks (OSPF)

- 10.0.0.192/30 — link to Active MLS
- 10.0.0.196/30 — link to Standby MLS
- 10.0.0.208/30 — link to Building 1
- 10.0.0.212/30 — link to Building 3

Network Address Translation (NAT)

Type	NAT Overload (PAT)
NAT Inside Interfaces	FastEthernet 0/0, FastEthernet 1/0
NAT Outside Interface	GigabitEthernet 9/0
Translation	All inside networks share the public IP of Gi9/0 (20.0.0.5) using overload

Inside Subnets Permitted for NAT (ACL 1):

- 10.0.0.144/28 (Floor 1)
- 10.0.0.176/29 (Floor 2)
- 10.0.0.160/28 (Floor 3)

```
access-list 1 permit 10.0.0.144 0.0.0.15
access-list 1 permit 10.0.0.176 0.0.0.7
access-list 1 permit 10.0.0.160 0.0.0.15
ip nat inside source list 1 interface GigabitEthernet9/0 overload

interface GigabitEthernet9/0
 ip nat outside
interface FastEthernet0/0
 ip nat inside
interface FastEthernet1/0
 ip nat inside
```

Building 3

Floor 1

Number of Hosts	8 PCs
VLAN	VLAN 70
Subnet	10.0.0.128/28
Gateway (Virtual IP)	10.0.0.129 (virtual — HSRP Active/Standby on Multilayer Switch)
Broadcast Address	10.0.0.143
DHCP	Assigned by the Multilayer Switch (Active)
Usable IP Addresses	14
Used IPs	8 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 11
Available IPs (not used)	3

Access Switch — Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports)

Interface Range	Mode	Purpose
Fa0/3 – Fa0/10	Access	Connects PCs — assigned to VLAN 70
Fa0/1 – Fa0/2	Trunk	Uplink to the Multilayer Switch
Fa0/11 – Fa0/24, Gi0/1 – Gi0/2	Unused	Reserved for future expansion

Floor 2

Number of Hosts	14 PCs
VLAN	VLAN 80
Subnet	10.0.0.32/27
Gateway (Virtual IP)	10.0.0.33 (virtual — HSRP Active/Standby on Multilayer Switch)
Broadcast Address	10.0.0.63
Usable IP Addresses	30
Used IPs	14 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 17
Available IPs (not used)	13

Access Switch — Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports)

Interface Range	Mode	Purpose
Fa0/3 – Fa0/16	Access	Connects PCs — assigned to VLAN 80
Fa0/1 – Fa0/2	Trunk	Uplink to the Multilayer Switch

Interface Range	Mode	Purpose
Fa0/17 – Fa0/24, Gi0/1 – Gi0/2	Unused	Reserved for future expansion

Floor 3

Number of Hosts	10 PCs
VLAN	VLAN 90
Subnet	10.0.0.112/28
Gateway (Virtual IP)	10.0.0.113 (virtual — HSRP Active/Standby on Multilayer Switch)
Broadcast Address	10.0.0.127
DHCP	Assigned by the Multilayer Switch (Active)
Usable IP Addresses	14
Used IPs	10 PCs + 3 reserved addresses (Active Gateway, Standby Gateway, Virtual IP) = 13
Available IPs (not used)	1

Access Switch — Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports)

Interface Range	Mode	Purpose
Fa0/3 – Fa0/12	Access	Connects PCs — assigned to VLAN 90
Fa0/1 – Fa0/2	Trunk	Uplink to the Multilayer Switch
Fa0/13 – Fa0/24, Gi0/1 – Gi0/2	Unused	Reserved for future expansion

Multilayer Switch 4 (Active)

Model	Cisco Catalyst 6500 (Multilayer Switch)
Role	Active switch in HSRP — provides routing between the floors of Building 3 and forwards traffic to the Edge Router.

VLANs and Interfaces

VLAN 70 (Floor 1)	Interface IP 10.0.0.130/28 · Virtual IP (HSRP) 10.0.0.129 — default gateway for Floor 1 PCs
VLAN 80 (Floor 2)	Interface IP 10.0.0.34/27 · Virtual IP (HSRP) 10.0.0.33 — default gateway for Floor 2 PCs
VLAN 90 (Floor 3)	Interface IP 10.0.0.114/28 · Virtual IP (HSRP) 10.0.0.113 — default gateway for Floor 3 PCs

Trunk Interfaces

Gi1/0/1 – Gi1/0/3 configured as trunk ports, connecting the Multilayer Switch to the access switches of the three floors.

DHCP Configuration

Pool F7 (Floor 1)	Network 10.0.0.128/28
Default Router	10.0.0.129
DNS	10.0.0.180
Excluded Addresses	10.0.0.129 – 10.0.0.131

Pool F8 (Floor 2)	Network 10.0.0.32/27
Default Router	10.0.0.33
DNS	10.0.0.180
Excluded Addresses	10.0.0.33 – 10.0.0.35

Pool F9 (Floor 3)	Network 10.0.0.112/28
Default Router	10.0.0.113
DNS	10.0.0.180
Excluded Addresses	10.0.0.113 – 10.0.0.115

Uplink to Edge Router

Interface Gi1/0/4 is configured as a routed port with IP address 10.0.0.201/30, connected directly to the Edge Router.

HSRP Configuration

- This Multilayer Switch is configured as the Active HSRP device for VLANs 70, 80, and 90.
- The second Multilayer Switch in Building 3 acts as the Standby HSRP device.
- Priority is set to 150 (higher than the default 100), ensuring this switch always takes the Active role when available.
- Preempt is enabled — if the Active switch fails and later recovers, it automatically reclaims the Active role.

Routing

- Default route configured to forward unknown traffic to the Edge Router (next-hop 10.0.0.202).
- OSPF enabled for internal routing between VLANs and to share routes with the Edge Router, ensuring connectivity between buildings.

Multilayer Switch 5 (Standby)

Model	Cisco Catalyst 6500 (Multilayer Switch)
Role	Standby switch in HSRP — provides redundancy for the Active switch and takes over only if it fails.

VLANs and Interfaces

VLAN 70 (Floor 1)	Interface IP 10.0.0.131/28 · Virtual IP (HSRP) 10.0.0.129 — default gateway for Floor 1 PCs
VLAN 80 (Floor 2)	Interface IP 10.0.0.35/27 · Virtual IP (HSRP) 10.0.0.33 — default gateway for Floor 2 PCs
VLAN 90 (Floor 3)	Interface IP 10.0.0.115/28 · Virtual IP (HSRP) 10.0.0.113 — default gateway for Floor 3 PCs

Trunk Interfaces

Gi1/0/1 – Gi1/0/3 configured as trunk ports, connecting the Multilayer Switch to the access switches of the three floors.

DHCP Configuration

Pool F7 (Floor 1)	Network 10.0.0.128/28
Default Router	10.0.0.129
DNS	10.0.0.180
Excluded Addresses	10.0.0.129 – 10.0.0.131

Pool F8 (Floor 2)	Network 10.0.0.32/27
Default Router	10.0.0.33
DNS	10.0.0.180
Excluded Addresses	10.0.0.33 – 10.0.0.35

Pool F9 (Floor 3)	Network 10.0.0.112/28
Default Router	10.0.0.113
DNS	10.0.0.180
Excluded Addresses	10.0.0.113 – 10.0.0.115

Uplink to Edge Router

Interface Gi1/0/4 is configured as a routed port with IP address 10.0.0.205/30, connected directly to the Edge Router.

HSRP Configuration

- This Multilayer Switch operates as the Standby HSRP device for VLANs 70, 80, and 90.
- No priority or preempt configured by default — it takes over only if the Active switch fails, and does not reclaim the Active role once the Active switch comes back online.

Routing

- Default route configured to forward unknown traffic to the Edge Router (next-hop 10.0.0.206).
- OSPF enabled to advertise VLAN networks and learn routes from the Edge Router, ensuring inter-building connectivity.

Edge Router 2

Model	PT-Router (Packet Tracer Default Router)
Role	Connects the internal network of Building 3 to the ISP and applies NAT. It also acts as the interconnection point between buildings, ensuring devices in Building 1, Building 2, and Building 3 can communicate through routing.

Interfaces and IP Addresses

Interface	IP Address	Connects To
FastEthernet 0/0	10.0.0.202/30	Active Multilayer Switch
FastEthernet 1/0	10.0.0.206/30	Standby Multilayer Switch
Serial 2/0	20.0.0.9/30	ISP Router
Serial 3/0	10.0.0.214/30	Edge Router of Building 2

Routing

- Default route: 0.0.0.0/0 → next-hop 20.0.0.10 (ISP).
- OSPF configured to advertise the routed links to the Multilayer Switches, allowing all internal subnets of the three floors to reach the router, and ensuring inter-building connectivity as well as access to the ISP.

Advertised Networks (OSPF)

- 10.0.0.200/30 — link to Active MLS
- 10.0.0.204/30 — link to Standby MLS
- 10.0.0.212/30 — link to Building 2

Network Address Translation (NAT)

Type	NAT Overload (PAT)
NAT Inside Interfaces	FastEthernet 0/0, FastEthernet 1/0
NAT Outside Interface	Serial 2/0
Translation	All inside networks share the public IP of Se2/0 (20.0.0.9) using overload

Inside Subnets Permitted for NAT (ACL 1):

- 10.0.0.128/28 (Floor 1)
- 10.0.0.32/27 (Floor 2)
- 10.0.0.112/28 (Floor 3)

```
access-list 1 permit 10.0.0.128 0.0.0.15
access-list 1 permit 10.0.0.32 0.0.0.31
access-list 1 permit 10.0.0.112 0.0.0.15
ip nat inside source list 1 interface Serial2/0 overload

interface Serial2/0
 ip nat outside
interface FastEthernet0/0
 ip nat inside
interface FastEthernet1/0
 ip nat inside
```

ISP Building

This building represents the Internet Service Provider (ISP). It provides external services (Web and FTP) that can be accessed from the enterprise network — Buildings 1, 2, and 3.

Network	192.168.1.0/24
Default Gateway	192.168.1.1 (configured on the PT-Router)

Server 1 — Web Server

IP Address	192.168.1.2/24
Service	HTTP (Web Browser)
Access	Accessible by clients from the enterprise network

Server 2 — FTP Server

IP Address	192.168.1.3/24
Service	FTP
Authentication	Username: said · Password: 12345

Access Switch — Cisco Catalyst 2960 (24 FastEthernet ports + 2 GigabitEthernet ports)

Interface Range	Mode	Purpose
Fa0/1 – Fa0/3	Access	Connects PCs — assigned to VLAN N/A (default)
—	Trunk	Uplink to the Multilayer Switch
Fa0/4 – Fa0/24, Gi0/1 – Gi0/2	Unused	Reserved for future expansion

ISP Router

Model	PT-Router (Packet Tracer Default Router)
-------	--

Interfaces and IP Addresses

Interface	IP Address	Connects To
FastEthernet 0/0	20.0.0.2/30	Edge Router of Building 1
FastEthernet 1/0	20.0.0.6/30	Edge Router of Building 2
Serial 2/0	20.0.0.10/30	Edge Router of Building 3
GigabitEthernet 9/0	192.168.1.0/24	ISP External Services Network

Network Address Translation (NAT)

Type	PAT — one-to-one public IP mapping per building
NAT Inside Interface	GigabitEthernet 9/0
NAT Outside Interfaces	FastEthernet 0/0, FastEthernet 1/0, Serial 2/0

Each building is mapped to a dedicated public IP when accessing the ISP:

- 20.0.0.2 → Building 1
- 20.0.0.6 → Building 2
- 20.0.0.10 → Building 3

This ensures that traffic from each building uses its own unique public IP. In case of a link failure, a building's traffic can be rerouted via another available public IP.

Static NAT (Port Forwarding)

- Web Server (192.168.1.2:80) accessible via 20.0.0.2:80, 20.0.0.6:80, and 20.0.0.10:80
- FTP Server (192.168.1.3:20–21) accessible via 20.0.0.2:20–21, 20.0.0.6:20–21, and 20.0.0.10:20–21

```
access-list 1 permit 192.168.1.0 0.0.0.255

interface GigabitEthernet9/0
 ip nat inside

interface FastEthernet0/0
 ip nat outside
 ip nat inside source list 1 interface FastEthernet0/0 overload
 ip nat inside source static tcp 192.168.1.2 80 20.0.0.2 80
 ip nat inside source static tcp 192.168.1.3 20 20.0.0.2 20
 ip nat inside source static tcp 192.168.1.3 21 20.0.0.2 21

interface FastEthernet1/0
 ip nat outside
 ip nat inside source list 1 interface FastEthernet1/0 overload
 ip nat inside source static tcp 192.168.1.2 80 20.0.0.6 80
 ip nat inside source static tcp 192.168.1.3 20 20.0.0.6 20
 ip nat inside source static tcp 192.168.1.3 21 20.0.0.6 21

interface Serial2/0
 ip nat outside
 ip nat inside source list 1 interface Serial2/0 overload
 ip nat inside source static tcp 192.168.1.2 80 20.0.0.10 80
 ip nat inside source static tcp 192.168.1.3 20 20.0.0.10 20
 ip nat inside source static tcp 192.168.1.3 21 20.0.0.10 21
```

Verification / Testing

Port Security

On interface Fa0/5, if another device is connected (e.g., replacing PC1), the port automatically shuts down and blocks the new device.

Access Control List (ACL)

PC0 (static IP 10.0.0.4) is not able to access the FTP server due to the applied ACL. However, PC0 can still access other services (Web, Email, DNS) without restriction.

HSRP Active/Standby Verification (Telnet)

Using Telnet to connect to the virtual IP (10.0.0.1) from any PC in Building 1, the session is always redirected to the Active Multilayer Switch — allowing quick verification of which switch is currently Active and which is Standby.

DNS Service

From any PC in the network, DNS resolves the domain `www.mahmoud.com` to the correct server IP.

Web Service

Using a web browser on any PC, the Web Server is accessible via its IP: 10.0.0.180.

FTP Service

From any PC in the topology, FTP access is possible using the server IP 10.0.0.180. For verification, a file named `net.txt` was uploaded to the FTP server from PC49 and successfully downloaded by PC56.

Email Service

Email functionality was tested between PC48 and PC55 (send/receive).

Inter-Building Connectivity

From any PC in Building 1, a successful ping was made to PCs in other buildings, confirming OSPF routing and connectivity.

NAT Verification

To test NAT, from any building an attempt was made to access an unknown external network. Each building goes out to the internet using its dedicated public IP — Building 1 → 20.0.0.2, Building 2 → 20.0.0.6, Building 3 → 20.0.0.10. Using a web browser or FTP client, external ISP services can be accessed via these public IPs.

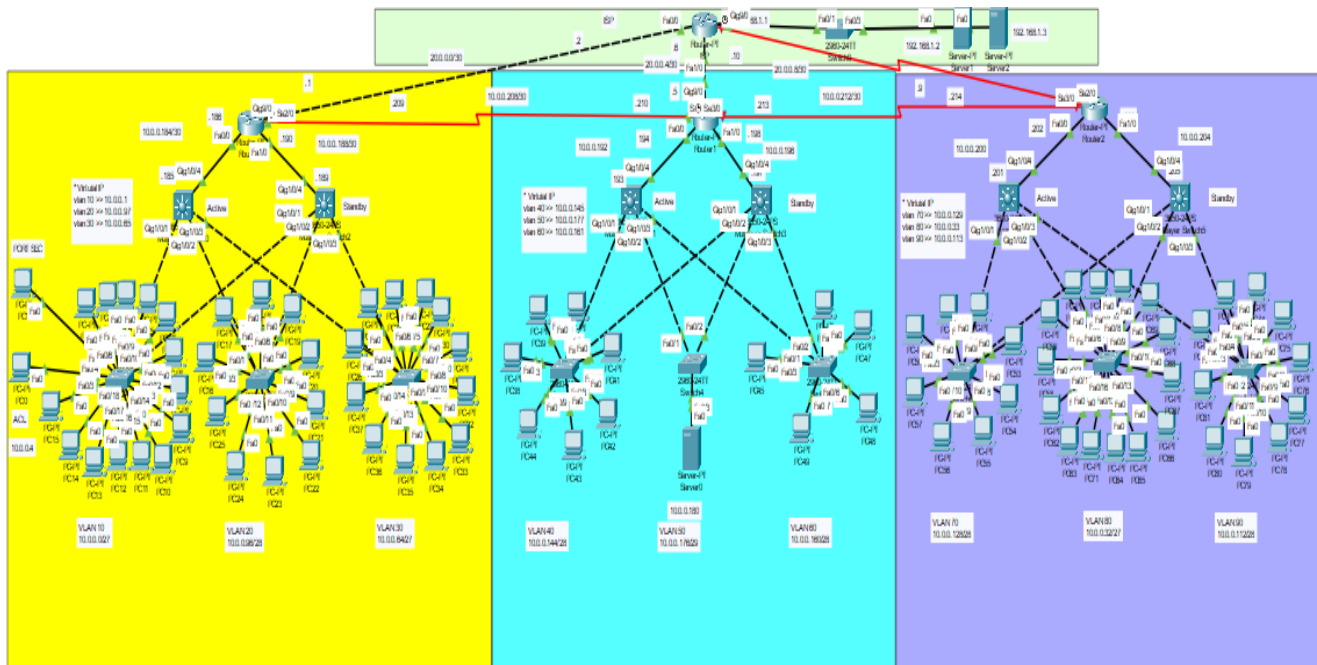
Conclusion

This project successfully demonstrates the design and implementation of a multi-building enterprise network. Key highlights include:

- VLAN segmentation and DHCP services per floor.
- Redundancy and load balancing with HSRP on multilayer switches.
- Secure access using Port Security and ACLs.
- Full inter-building connectivity via OSPF.
- External connectivity to the ISP through NAT (PAT) and dedicated public IPs.
- Hosting and accessing services (Web, DNS, FTP, Email) on ISP servers.

Through step-by-step testing, all network functionalities were verified, ensuring reliability, redundancy, and scalability of the design.

Topology Screenshot



Full network topology — Buildings 1, 2, 3 and ISP (Cisco Packet Tracer)